Exploratory Analysis of Breast Cancer Datasets

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Introduction

🧹 Data Selection and Cleaning

This section describes the datasets used in the analysis, their structure, origin, and the preprocessing steps applied to ensure data quality, consistency, and usability.

📊 Dataset 1: Breast Cancer Wisconsin (Original – Categorical)

📌 Overview

The **Breast Cancer Wisconsin (Original)** dataset contains clinical observations collected by Dr. William H. Wolberg. The data was gathered over time, resulting in natural chronological groupings.

1.1 Data Collection Groups

Group	Instances	Date
Group 1	367	January 1989
Group 2	70	October 1989
Group 3	31	February 1990
Group 4	17	April 1990
Group 5	48	August 1990
Group 6	49	January 1991 (updated)
Group 7	31	June 1991
Group 8	86	November 1991

Total: 699 samples (as of July 15, 1992)

> Note: Group 1 originally contained 369 samples; 2 were later removed.

1.2 Features Description 🧬

Feature	Categorical Values
Clump Thickness	1 – 10
Uniformity of Cell Size	1 – 10
Uniformity of Cell Shape	1 – 10
Marginal Adhesion	1 – 10
Single Epithelial Cell Size	1 – 10
Bare Nuclei	1 – 10
Bland Chromatin	1 – 10

Feature	Categorical Values
Normal Nucleoli	1 – 10
Mitoses	1 – 10
Class	**Benign / Malignant**

1.3 🌐 Data Sources

- https://www.kaggle.com/datasets/mariolisboa/breast-cancer-wisconsin-original-data-set
- https://archive.ics.uci.edu/dataset/15/breast+cancer+wisconsin+original

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1.4 🧹 Data Cleaning

- **Handling Missing Values**
    - The *Bare Nuclei* column contained non-numeric values (e.g., "?")
    - Converted to numeric and handled missing values appropriately
- **Target Variable Standardization**
    - Converted class labels:
        - `2 → Benign`
        - `4 → Malignant`
- **Identifier Issues**
    - The *Sample Code Number* was found to be non-unique
    - Replaced with a reliable index based on the DataFrame
- **Data Type Consistency**
    - Ensured all feature columns are numeric and within expected ranges

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📊 Dataset 2: Breast Cancer Wisconsin (Diagnostic – Numerical)

📌 Overview

This dataset contains features computed from digitized images of Fine Needle Aspirate (FNA) of breast masses. The features describe geometric and statistical properties of cell nuclei.

2.1 Feature Characteristics 🧬

Feature	Description
Radius	Mean distance from center to perimeter points
Texture	Standard deviation of gray-scale values
Perimeter	Perimeter of the nucleus
Area	Area of the nucleus
Smoothness	Local variation in radius lengths
Compactness	$(\text{Perimeter}^2 / \text{Area}) - 1.0$
Concavity	Severity of concave portions of the contour
Concave Points	Number of concave portions of the contour

Feature	Description
Symmetry	Symmetry of the nucleus
Fractal Dimension	"Coastline approximation" − 1
Reference:	
K. P. Bennett & O. L. Mangasarian (1992)	

2.2 🌐 Data Source

- https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

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2.3 📈 Dataset Summary

- **Total Columns:** 32
- **Target Variable:** Diagnosis
    - M → Malignant
    - B → Benign
- **Missing Values:** None detected
- **Index Column:** ID / Sample code number

2.4 🧹 Data Cleaning

- **Removed Irrelevant Columns**
    - Dropped column: `Unnamed: 32` (entirely null)
- **Target Variable Standardization**
    - Converted:
        - `M → Malignant`

        - `B → Benign`
- **Column Naming Consistency**
    - Ensured clear and standardized column names

📊 Dataset 3: Breast Cancer Survival (METABRIC)

📌 Overview

The METABRIC (Molecular Taxonomy of Breast Cancer International Consortium) dataset contains clinical and genomic data for breast cancer patients, focusing on survival outcomes.

3.1 🌐 Data Source

- https://www.kaggle.com/datasets/gunesevitan/breast-cancer-metabric

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3.2 🧬 Dataset Characteristics

- Includes:
    - Clinical variables (age, tumor size, etc.)
    - Gene expression data
    - Survival information
- Suitable for:
    - Survival analysis
    - Risk prediction
    - Advanced statistical modeling

3.3 🧹 Data Cleaning

- **Missing Value Handling**
    - Identified null values across multiple columns
    - Removed or filtered incomplete records
- **Data Validation**
    - Verified consistency across clinical variables
    - Checked distributions and outliers

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4. 🗄️ Final Data Integration

After cleaning and preprocessing all datasets:

- All datasets were standardized in terms of:
    - Naming conventions
    - Target labels
    - Data types
- A unified database was created:

```
breast_cancer_analysis.db
```

🔗 Integration Steps

- Loaded cleaned datasets into structured tables
- Ensured compatibility across schemas
- Enabled cross-dataset querying and analysis

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5. Exploratory Data Analysis

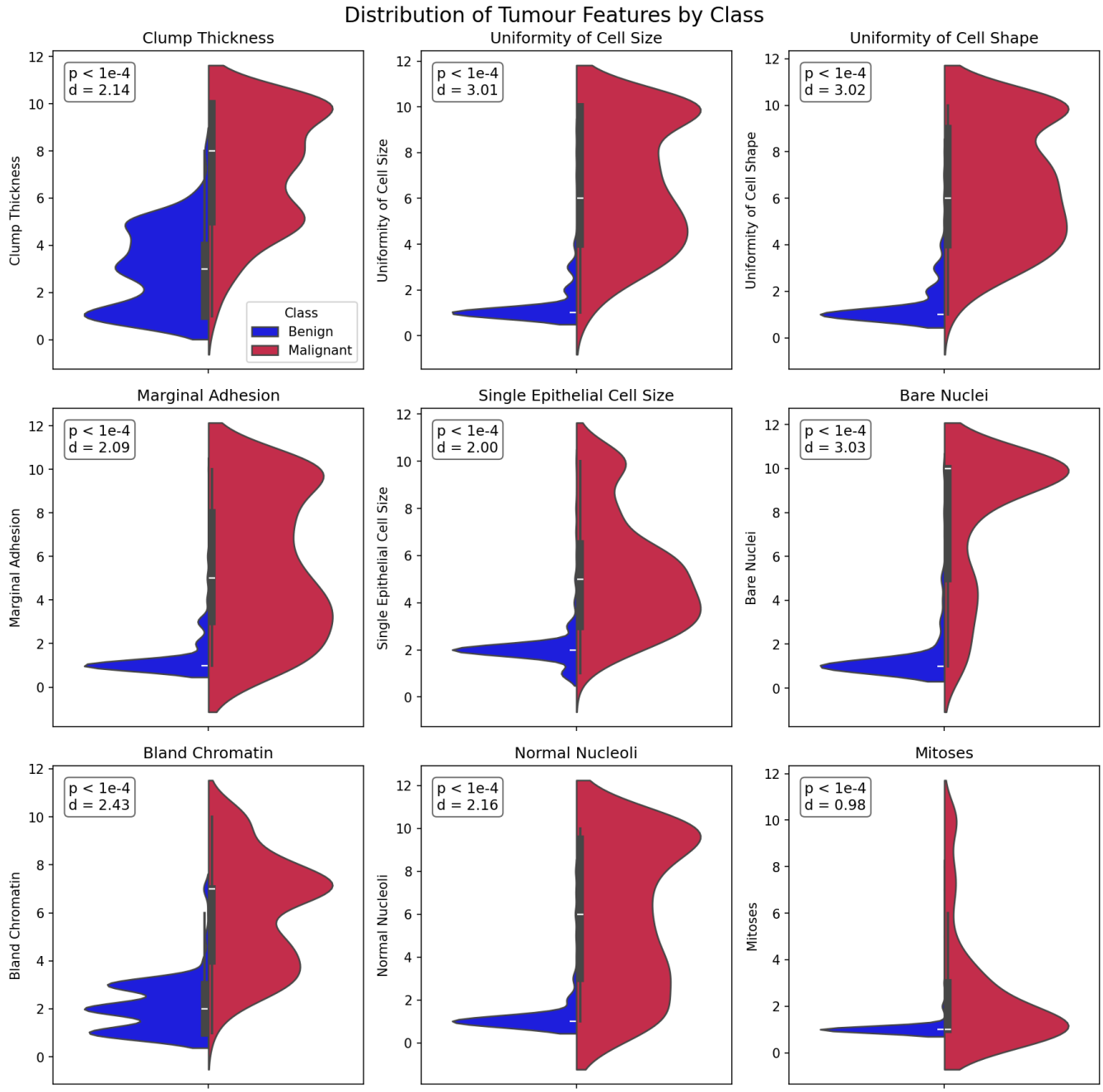

Figure 1: KDE plot of categorical breast cancer data

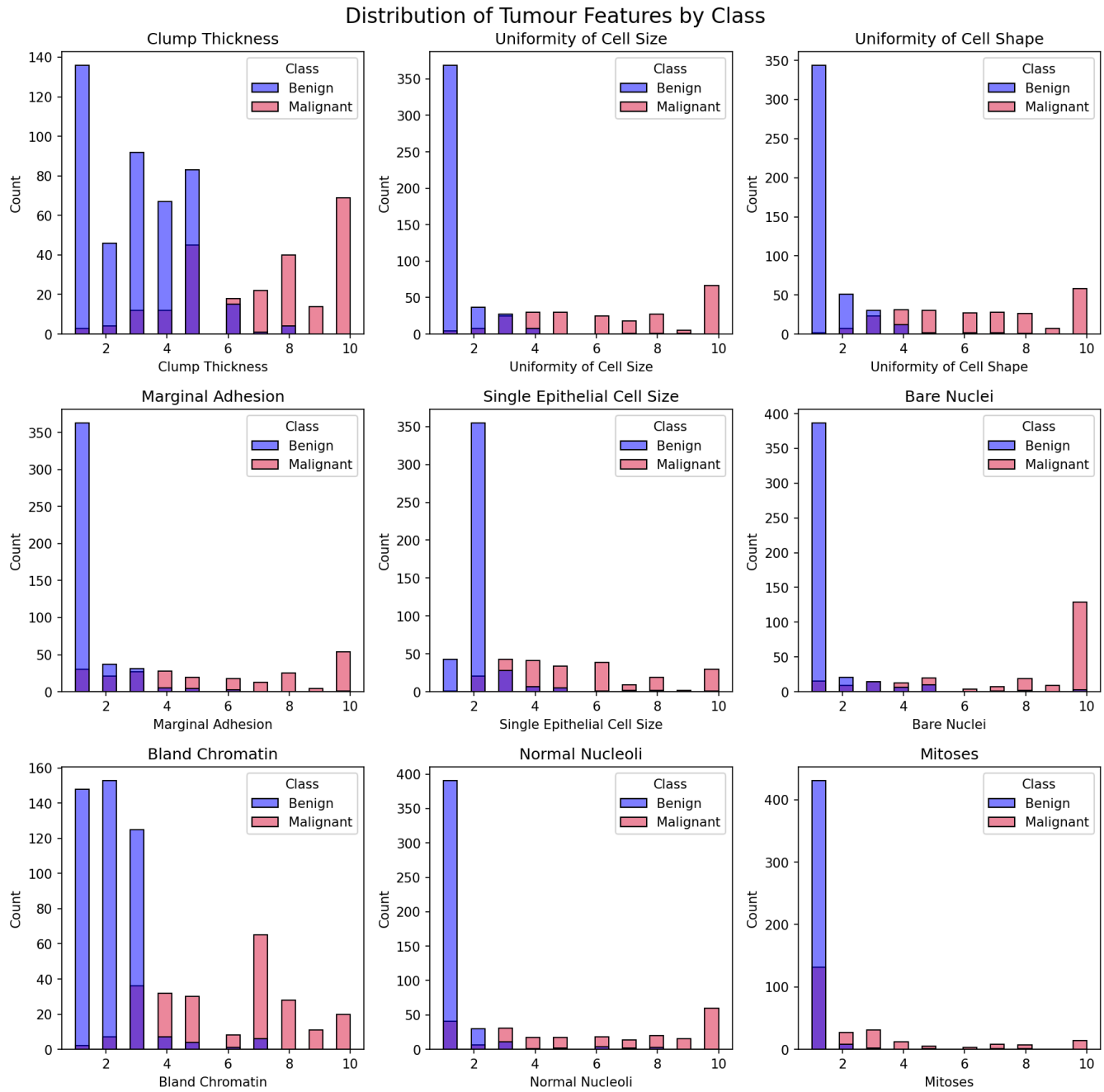

Figure 2: Histograms of categorical breast cancer data

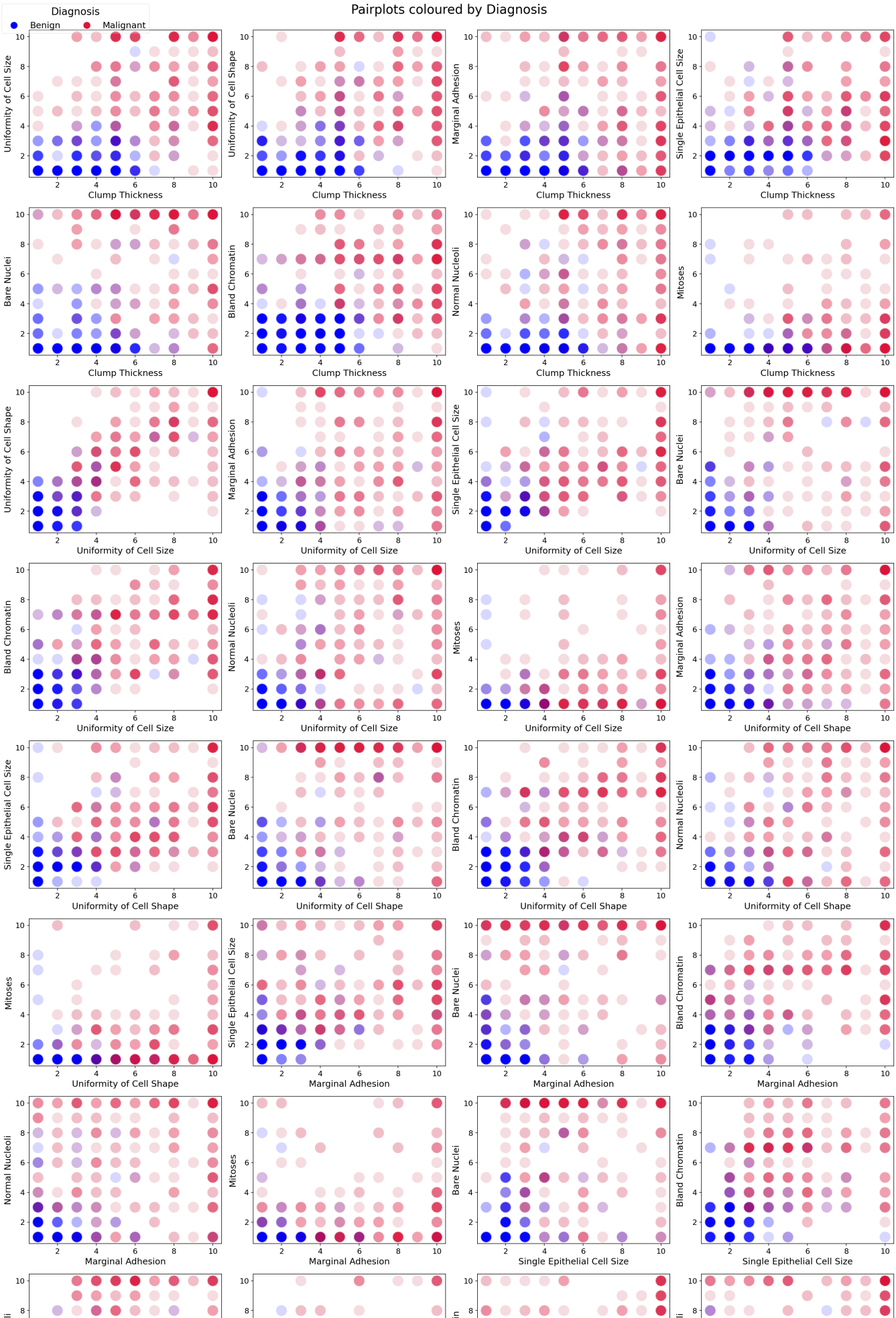
Pairplots coloured by Diagnosis
Diagnosis
Benign
Malignant
Uniformity of Cell Size
Uniformity of Cell Shape
Marginal Adhesion
Single Epithelial Cell Size
Bare Nuclei
Bland Chromatin
Normal Nucleoli
Mitoses
Clump Thickness

Figure 3: Pair plots of categorical breast cancer data